

COMP1702	Big Data	Faculty Header ID	Contribution: 100% of course
Course Leader: Hai Huang	Coursework		Deadline Date: 25 March 2026 (5 PM, UK time)
Feedback: Feedback and grades for the main sit will be released on 21 April (normally released within 17 working days of the original deadline, excluding weekends and university closure days). For the late sit, feedback and grades are normally released within 27 working days of the original deadline.			
Student Workload: This coursework is expected to take an average student who is up to date with tutorial work approximately 50 hours to complete.			
Learning Outcomes: 1. Explain the concept of Big Data and its importance in a modern economy. 2. Explain the core architecture and algorithms underpinning big data processing. 3. Analyse and visualize large data sets using a range of statistical and big data technologies. 4. Critically evaluate, select and employ appropriate tools and technologies for the development of big data applications.			
Plagiarism and AI declaration: - Guidance to the AI usage and the university policy and student declaration: guidance to AI usage; declaration of AI use. - Plagiarism is presenting somebody else's work as your own. It includes copying information directly from the Web or books without referencing the material; submitting joint coursework as an individual effort; copying another student's coursework; stealing coursework from another student and submitting it as your own work. Suspected plagiarism will be investigated and if found to have occurred will be dealt with according to the procedures set down by the University. - All material copied or amended from any source (e.g. internet, books) must be referenced correctly according to the reference style you are using. - Your work will be submitted for plagiarism checking. Any attempt to bypass our plagiarism detection systems will be treated as a severe Assessment Offence.			

Coursework Submission Requirements

- The last version you upload will be the one that is marked.
- An electronic copy of your work for this coursework must be fully uploaded on **the Deadline Date of 25 March 2026** using the link on the **coursework Moodle page for COMP1702**.
- For this coursework **you must submit a single report in PDF format**. In general, any text in the document must not be an image (i.e. must not be scanned) and would normally be generated from other documents (e.g. MS Office using "Save As .. PDF"). An exception to this is handwritten

mathematical notation, but when scanning do ensure the file size is not excessive.

- There are limits on the file size (see the relevant course Moodle page).
- Make sure that any files you upload are virus-free and not protected by a password or corrupted otherwise they will be treated as null submissions.
- Your work will not be printed in colour. Please ensure that any pages with colour are acceptable when printed in Black and White.
- You must NOT submit a paper copy of this coursework.
- All coursework must be submitted as above. Under no circumstances can they be accepted by academic staff.

The University website has details of the current Coursework Regulations, including details of penalties for late submission, procedures for Extenuating Circumstances, and penalties for Assessment Offences. See <http://www2.gre.ac.uk/current-students/regs>

Detailed Specification

You are expected to work **individually** and complete a report that addresses the following tasks. You need to cite all sources you rely on with **in-text style, Harvard format**. You can include material discussed in the lectures or labs, but independent research might be needed.

Task A (10 marks). Characteristics of Big Data

Please use real-world examples to explain the four characteristics of Big Data. Citations should be placed as close as possible to the borrowed information.

- **Word count:** 200 words $\pm 10\%$
- **Marking scheme (10 marks):** 7 for completeness & clarity of description; 3 for citations/references

Task B (40 marks). Hive Data Warehouse Design

Independently design and build a simple data warehouse using Apache Hive on the Hadoop virtual machine (VM) provided to you. The data warehouse must include exactly **3** tables with a total of at least **50** records across all tables. Then, create and execute **8** diverse HiveQL queries to demonstrate data retrieval/manipulation/analysis. Ensure the tables, data, and queries exhibit variety in structure (e.g., different data types) and complexity (e.g., including joins, aggregations, and functions). All work must be performed directly in your Hadoop VM environment—no online Hive services, cloud platforms, or external tools are allowed.

Your solution should clearly demonstrate your Hive data warehouse implementation. Include detailed explanations, discussions, and high-quality screenshots for every major step of the process (table design, building tables, populating tables, executing queries and showing query results).

For each HiveQL query, provide a clear, detailed explanation to query propose and the functions/operations involved. This explanation must appear immediately after the corresponding query code and screenshot(s). The screenshot(s) must prove that the query was successfully executed in Hive and produced the results.

- **Marking scheme (40 marks):**

- a) **Table design (5 marks):** 3 for design correctness & complexity; 2 for clarity of description.
- b) **Building & populating tables (15 marks):** 10 for completeness of evidence and clarity of description; 5 for design correctness.
- c) **Query design (20 marks):** 10 for completeness of evidence and clarity of description; 10 for query correctness & complexity.

Task C (30 marks). MapReduce Programming

Data: Suppose that there is a big file including student records of course grades from year 2000 to year 2026. The file contains billions of lines and is stored in a Hadoop cluster. Each line of this file contains information: (*Student_Id*, *Gender*, *Course_Id*, *Course_Grade*, *School*, *Year*). Example data is as follows:

SC001238, male, COMP1702, 65, CMS school, 2019
SM100235, female, ECON1011, 77, BS school, 2017
.....

Task: You need to select **one** of the tasks based on the last digit of your student ID number.

Task 1: Output the average grade of each course for the year 2021.

Task 2: Output the minimum grade for each course for the year 2022.

Task 3: Output the schools and their student counts for the year 2023.

Task 4: Output the number of female students for the year 2024.

Task 5: Output the number of courses for each year.

If the last digit of your student ID number

$$d = \begin{cases} 0 \text{ or } 1, & \text{please select Task 1} \\ 2 \text{ or } 3, & \text{please select Task 2} \\ 4 \text{ or } 5, & \text{please select Task 3} \\ 6 \text{ or } 7, & \text{please select Task 4} \\ 8 \text{ or } 9, & \text{please select Task 5} \\ \text{other,} & \text{please select Task 5} \end{cases}$$

For example, if your student id number is: “011340894”, the last digit is 4. Then you need to do task 3. **Please ignore any number after “–” in your student id.** For example, if your student id shows “011340894 – 5”, please ignore 5. The digit “4” is the last digit and you need to select task 3.

Please highlight your student ID number and the task you need to solve in your solution. **An incorrect selection will result in zero marks for this task.** If you are not sure which task to be selected, please contact your tutor.

Task requirements: Please implement a MapReduce algorithm by writing **pseudo-code (only pseudo-code is acceptable)** to solve the task assigned to you. Please follow the way of writing pseudo-codes introduced in the lecture of “Advanced MapReduce Programming”. Your design is expected to be as efficient as possible.

You should also explain how the input is mapped into (key, value) pairs by the map stage, i.e., specify what is the key and what is the associated value in each pair, and, how the key(s) and value(s) are computed. Then you should explain how the output (key, value) pairs of the map stage are processed by the reduce stage to get the final answer(s). You need to discuss the efficiency of your algorithm (How your design can make your algorithm efficient).

- **Note:** All pseudo-code must be included as plain text directly in the report. Providing screenshots or images of pseudocode is not permitted and will result in zero marks.
- **Marking scheme (30 marks):** 15 for design correctness & pseudocode clarity; 15 for clarity of algorithm explanation & performance analysis.

Task D (20 marks). Big Data Project Analysis

The PA company is a leading provider of precision agriculture service. Precision agriculture is the science of gathering, processing, and analysing temporal, spatial and individual data. It combines other information to support management decisions according to estimated variability for improved resource use efficiency, productivity, quality, profitability.

The PA company is now plan to develop a big data project to meet the following requirements: help worldwide users better understanding the implications of the weather and making contingency plans; buying supplies, such as fertilizer and seeds; as well as maintaining and monitoring the quality of yield, whether livestock or crops; knowing the variety of cultivated plants, conditions of its growth and its needs of seeds; choosing the type of fertilizer and pesticides, understanding their employment conditions and their impact on the climate- soil-plant; recognizing daily water needs for each kind of plant; calculating the median and mean values of yield; studying the conditions of natural environment; estimating the financial revenue and manage the potential risks.

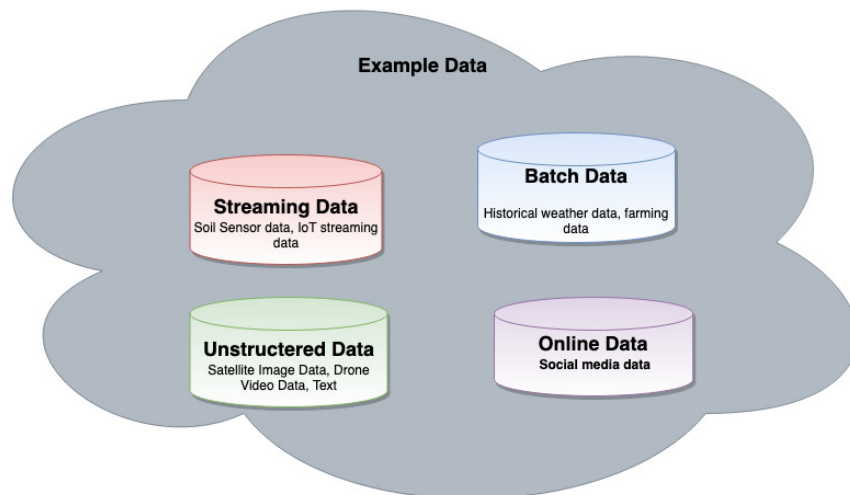


Figure 1. Example data of PA company.

Task D.1: The volume of big data is expected to be more than 500 Petabytes. The data will come from various sensors, satellites, drones, social media, market data,

Online news feed etc. Figure 1 shows some example data of PA company. Some IT technician planned to build a data warehouse to store data for further data analysis tasks, but some others believed data lake is a better choice. Which choice do you prefer? Please justify your choice with technical detail.

- **Word count:** 300 words $\pm 10\%$
- **Marking scheme (10 marks):** 4 for correctness of choice; 4 for clarity of explanation; 2 for citations/references.

Task D.2: Some prediction and analytics services provided by the PA company require to response in a few seconds after the arrival of new data. Namely, they are real time or near real time prediction and analytics tasks. Some IT managers suggested a popular distributed processing framework — MapReduce to implement these tasks. Do you agree with that? Do you have another choice? Please justify your choice.

- **Word count:** 300 words $\pm 10\%$
- **Marking scheme (10 marks):** 4 for correctness of choice; 4 for clarity of explanation; 2 for citations/references.